Achieving Net-Zero for Artificial Intelligence: Big-Tech's Sustainability Challenges from 2030 to 2050

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Abstract

Today big-tech companies—the largest, most financially successful, and most influential firms developing technology platforms and products— are responsible for 40% of overall data center energy consumption, and they pose a significant risk to environmental sustainability. To mitigate this risk, many of these corporations are claiming to achieve net-zero carbon emissions by 2030, including major big-tech companies, such as Amazon, Google, Meta and Microsoft. However, carbon accounting methodologies can be misleading, due to various strategies employed such as carbon offsetting. In this article, we analyze the carbon emissions of four big-tech companies and project their potential carbon emissions from 2030 to 2050. We provide an in-depth look at their carbon offsetting strategies and quantify their true environmental impact. Furthermore, we project their AI-driven energy consumption and show that there is a pressing need to tackle sustainability challenges within the next five years to mitigate the environmental impact of data centers and AI.

Keywords: Sustainability, data centers, AI, cloud, carbon offsetting, net-zero emissions, big-tech

1. Introduction

Reaching net-zero carbon targets has become a crucial goal for many big-tech companies, a term meant to describe the largest, most financially successful, and most influential firms developing technology platforms and products [1, 2]. With increased societal and regulatory pressure to mitigate the impacts of climate change [3, 4], big-tech companies have announced efforts to fully decarbonize their supply chains [5]. However, there are several challenges to achieving net-zero goals without resorting to use of carbon offsetting[1] strategies. The complex supply chain, high energy consumption, and utilization of non-renewable resources are the main culprits of the high carbon emissions of the big-tech companies. In terms of energy usage, some big-tech companies such as Google or Amazon now utilize more energy than some of the developed countries [6]. With the rise of artificial intelligence (AI), there is even a growing need to use more electricity than ever to operate. From transitioning to renewable energy sources and enhancing energy efficiency to managing carbon offsets and addressing indirect emissions, big-tech companies must address significant technical, financial, and operational challenges.

One area of extremely large energy consumption within big tech is their data center businesses. Data centers, which are the backbone of modern digital infrastructure, require vast amounts of electricity to operate. Although most companies strategize to address their increasing energy consumption, the overall demand for electricity continues to grow as the need for data center services is increasing and even becoming a bigger challenge due to boom in use of AI services such as ChatGPT [7, 8]. The rapid growth of AI is expected to be a major challenge in near future, as these applications require even more computational power.

[1] Carbon offsetting refers to the practice of compensating for carbon dioxide emissions by funding equivalent carbon dioxide savings elsewhere, such as through renewable energy projects, forest conservation, or other greenhouse gas reduction initiatives. (See Section 3 for more details.)

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One of the strategies that big-tech companies utilize is to increase the use of renewable energy sources to reduce their environmental impact. However, the intermittent nature of renewable energy requires substantial investment in energy storage solutions, such as batteries, to ensure a consistent power supply. Additionally, the geographic distribution of renewable energy resources may not align with the locations of data centers, necessitating further infrastructure development and investment [9]. On the other hand, carbon offsets have become a popular tool for big-tech companies aiming to achieve net-zero targets, but relying heavily on offsets can be problematic [10]. The effectiveness of carbon offsets can vary widely, and there is often a lack of transparency and accountability in the offset market. Ensuring that offsets represent real, permanent, and additional carbon reductions is crucial, but verifying these qualities can be difficult. Furthermore, offsets do not address the root cause of emissions and can sometimes serve as a distraction from more direct mitigation efforts. In this article, we explore and quantify the challenges big-tech companies face towards net-zero carbon emissions. We aim to provide a forward-looking outlook of the data center industry, which is heavily impacted by hyperscalers, such as Amazon, Google, Meta and Microsoft, as well as to quantify the impact of AI on carbon emissions of data centers.

2. Gaps in Carbon Accounting and Reporting for Big Tech

The energy consumption of big tech companies has become one of the major focuses on achieving sustainability and climate goals. As these companies grow and expand their digital infrastructure, their carbon footprints have come under intense scrutiny. Companies like Google, Apple, and Microsoft have made significant efforts in reducing their carbon footprints by investing in renewable energy and improving energy efficiency. For instance, Google has been claiming to be carbon neutral since 2007 and aims to run all its data centers on carbon-free energy by 2030 [11]. Apple claims to have achieved carbon neutrality for its global corporate operations and is working towards making its entire supply chain and product lifecycle carbon neutral by 2030 [12]. Microsoft has committed to becoming carbon negative by 2030 and aims to remove all the carbon it has emitted since its founding by 2050 [13]. Despite these efforts, the overall carbon footprint of the tech industry remains substantial, with many companies still heavily rely on fossil fuels and facing various challenges to manage indirect emissions.

While energy efficiency of the computing hardware and data center cooling solutions have increased over the years, the exponential growth in use of data center resources, which is driven by AI, continues to increase the overall energy demand [14]. One of the critical issues in evaluating the carbon footprint of big tech companies is the lack of comprehensive and accurate data. Missing data points can significantly hinder the ability to perform accurate sustainability analysis and generate reliable reports. For example, big-tech companies do not usually provide detailed information on energy consumption at different stages of supply chain of their manufacturing processes, emissions from third-party services, such as logistics and transportation, and the lifecycle emissions of products is often incomplete or unavailable. This lack of data can lead to underestimations or overestimations of the true carbon footprint, making it difficult to set realistic targets and track progress effectively. Furthermore, inconsistencies in reporting standards and methodologies across companies complicate comparisons and benchmarking.

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Standardized reporting frameworks and methodologies, such as the Greenhouse Gas Protocol, can help ensure consistency and comparability across companies [15]. Adopting such frameworks facilitate benchmarking and the sharing of best practices, driving industry-wide improvements in sustainability performance. Collaboration among companies, industry groups, and regulatory bodies can further enhance the quality and availability of data. By working together, stakeholders can develop common standards, share data and insights, and promote the adoption of best practices.

Nevertheless, the current state of carbon emissions, while showing some progress, highlights the need for continued efforts to reduce the industry's carbon footprint. Accurate and comprehensive data collection is essential for effective sustainability analysis and reporting, enabling companies to target their efforts, comply with regulations, and build trust with stakeholders. But the current status of data transparency is low. Data center operators and hardware vendors often hesitate to share detailed data about their operational and embodied carbon emissions for a variety of reasons. One key concern is that, as mission-critical infrastructure, data centers may become vulnerable if too much information is disclosed, potentially attracting unwanted attention from cybercriminals and other malicious actors [16]. Additionally, many private companies believe that revealing specific details about their infrastructure, including the types of hardware they employ and their energy efficiency strategies, could compromise their competitive advantage in the marketplace. Therefore, it is essential to propose methods to predict the carbon emissions by using the available data from sustainability reports of big-tech companies. In this study, we aim to provide a more realistic picture of carbon emissions of big-tech companies by taking into account the carbon offsetting strategies employed to reduce carbon emissions.

3. Assessing the Historical Carbon Footprint of Big Tech Companies

Companies such as Google, Apple, Microsoft, Amazon, and Facebook have made significant commitments to sustainability, yet their carbon footprints remain substantial due to the vast scale of their operations. Analyzing these footprints reveals the real challenges of these companies. In Table 1, we show carbon emissions of big-tech companies from 2020 to 2023 [5, 11, 13, 17]. Google, Meta and Microsoft all report over 10% year-over-year (YoY) increase, while Amazon shows only 3% average YoY increase in their carbon emissions.

One reason for Amazon’s lower carbon emissions is rooted in the way they report carbon emissions. Starting from 2022, Amazon updated their carbon footprint by incorporating emission reduction efforts, like renewable energy purchases and improvements in transportation efficiency [5], which is the main reason for their lower YoY increase in their carbon emissions. Some recent studies argue actual carbon emissions from the sustainability reports of big tech companies are vastly different than what is being reported [18]. In fact, some sources claim that the actual carbon emissions are 6 to 7x higher than the reported carbon emissions [19]. On the other hand, a carbon emission increase higher than 10% indicates that the carbon emissions are growing at a higher rate than the projections, which is mainly due to the boom in AI services [20, 21].

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	Carbon Emissions (mmt CO2-eq)				
	2020	2021	2022	2023	Increase-per-year (%)
Amazon	60.64	71.54	70.74	68.82	3%
Google	8.56	10.77	12.61	14.31	17%
Meta	5.09	5.77	8.46	7.44	12%
Microsoft	12.2	14.39	16.53	17.16	10%

Table 1: Carbon emissions of four major big-tech companies from 2020 to 2023.

As tech companies strive for net-zero carbon emissions, they often rely on a combination of carbon offsetting and power purchase agreements (PPAs) to achieve their goals. Carbon offsetting involves compensating for emissions by investing in projects that reduce or remove carbon dioxide from the atmosphere. These projects can include reforestation, renewable energy development, and methane capture from landfills. While carbon offsetting can help companies achieve net-zero targets, it has faced criticism. One of the leading non-profit organizations in carbon emissions "Science Based Targets initiative" (SBTi) suggests that the carbon accounting methodologies are significantly skewed to paint a brighter picture than the actual from the environmental impact perspective [22]. For instance, many companies use renewable energy credits (RECs) or carbon offsets to "neutralize" their emissions, but these credits don't always represent actual reductions. For example, if a company purchases RECs for renewable energy, it may still use non-renewable sources in practice while claiming lower emissions on paper. As a result, companies have started to take action to update their carbon footprint methodologies. As an example, Google no longer claims to be net-zero carbon and pledge to invest in renewable energy at a higher rate, rather than purchasing carbon credits [22].

Despite these challenges, carbon offsetting remains a valuable tool for tech companies to address emissions that are difficult to eliminate entirely. Companies like Microsoft have committed to using carbon offsets to address their residual emissions and have invested in innovative carbon removal technologies. In addition to carbon offsetting, big-tech companies employ PPAs, which are long-term contracts between a company and a renewable energy provider, to address their sustainability challenges. Under a PPA, a company agrees to purchase a specified amount of renewable energy at a fixed price, providing financial certainty for the energy provider and ensuring a steady supply of green energy for the company. PPAs have become a popular strategy for tech companies to reduce their Scope 2 emissions, which covers emissions associated with the production of purchased energy, and support the growth of renewable energy. Despite these challenges, many tech companies have successfully implemented PPAs as part of their sustainability strategies. Google, for example, has signed multiple PPAs to purchase renewable energy for its data centers, contributing to its goal of running all its operations on carbon-free energy by 2030 [11].

4. Predicting 2030-2050 AI-related Energy Demand and Carbon Emissions

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To estimate the AI-related energy consumption, we utilize projections from Evernote [23], where they use CAGR for NVIDIA GPU sales and NVIDIA's GPU market share. Since the projections are limited to until 2028, we use logarithmic fit to estimate the AI power demand up to 2050, as AI power demand is expected to saturate over time [21, 23]. To estimate the total data center energy consumption, we use the 10% YoY high growth scenario from EPRI [24], which assumes the rate of growth in data center investments will continue in a similar trend until 2030. In Figure 4, we show global total energy consumption projections until 2050 for AI and data centers (left) and the percentage of AI-driven energy consumptions (right). Various sources estimate AI-driven energy consumption to significantly increase in the next 5 years, then slow down and saturate eventually [21]. As Figure 1 shows, the impact of AI-driven energy consumption will peak at around 2028, accounting for over 60% of the overall energy consumption of data centers across the world. Therefore, there is an urgency to tackle the impact of AI in coming years. Overall, data center energy consumption is expected to remain as one of the major challenges against sustainable computing, which urges big-tech companies to take major steps to mitigate.

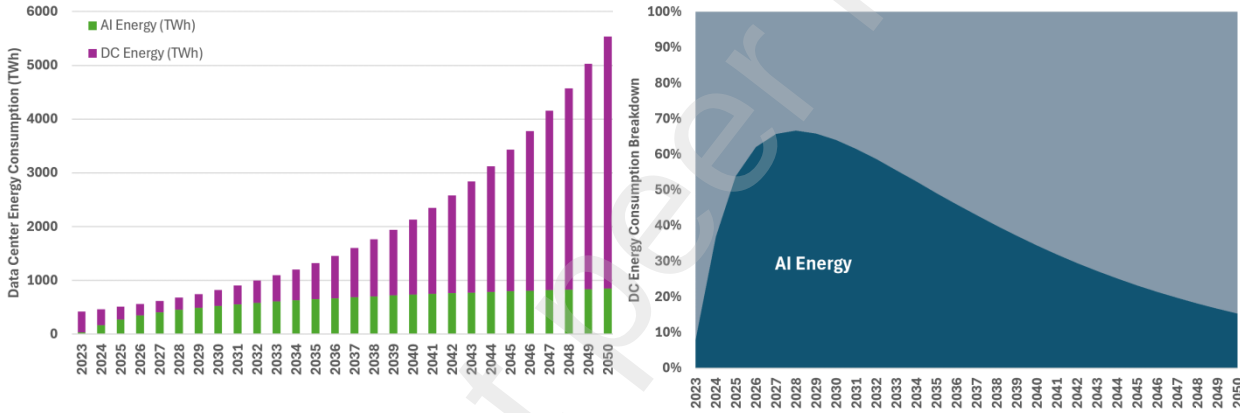

Figure 1: Total data center and AI-driven energy consumption projections (left) and percentage of AI-driven power consumption compared to total data center energy consumption (right).

To project the carbon footprint of big-tech companies, namely Amazon, Google, Meta, and Microsoft, we utilize the CAGR (compound annual growth rate) of each company and estimate the business growth. We project the future carbon emissions based on recently released carbon emission numbers from their sustainability reports. In Figure 2, we show the projected carbon emissions of Amazon from 2030 to 2050. Amazon has the highest carbon emissions among other big-tech companies we study in this article. Furthermore, Amazon sustainability reports indicate that they use the highest percent of carbon offsetting. As Figure shows, Amazon's carbon accounting strategy of their Scope 2 emissions is significantly different than the actual carbon emissions when not using any carbon offsetting. The projected carbon emissions of Amazon is 56% higher than the reported numbers in their sustainability report, when we remove the purchased carbon credits from their actual carbon emissions. Similar trends hold true for other big-tech companies we study in this article at different degrees. This result shows the significant need for transparent reporting of carbon emissions to tackle the sustainability issues in data center industry.

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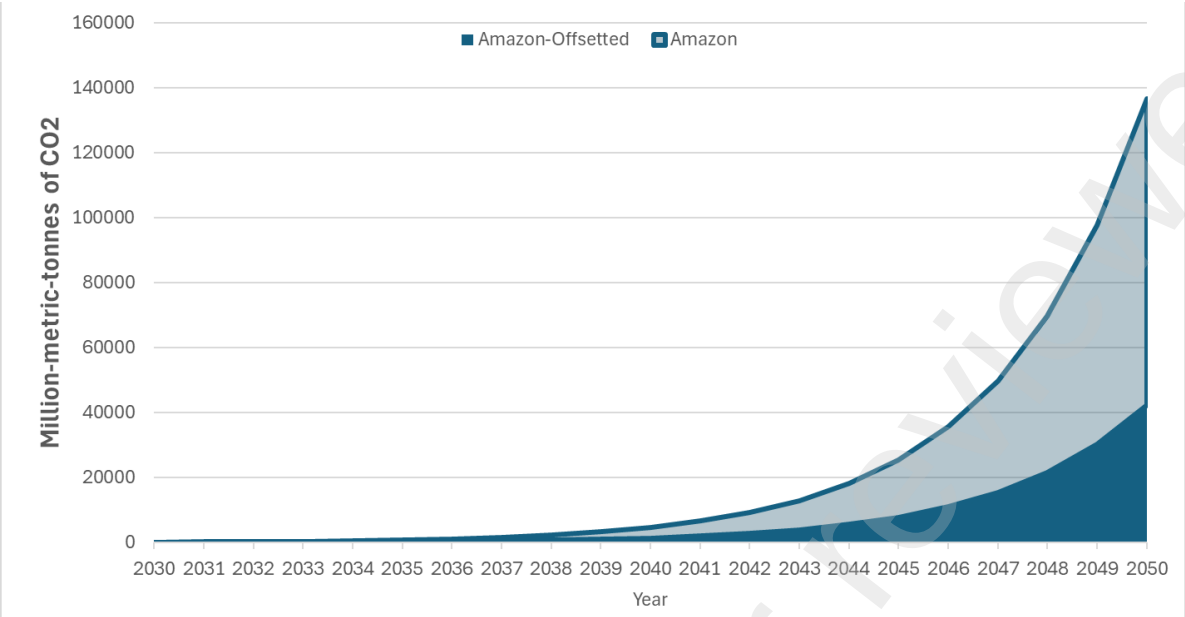

Figure 2: Amazon carbon emissions with and without carbon offsetting.

In addition to baseline growth scenarios, we also evaluate various cases such as carbon offsetting percentages, growth of renewable energy resources, AI-driven energy efficiency improvements and software-driven improvements.

4.1. Sustainability solutions for Big Tech and AI

Multiple opportunities exist to make Big Tech and AI more sustainable. The first is renewable electricity supply. In a recent report, U.S. Energy Information Administration (EIA) projects that the renewables will account for 44% of the electricity production by 2050, up from 21% from 2021. To estimate the impact of transition to renewable electricity on big-tech carbon emissions, we first need to quantify the impact of this transition, in terms of carbon intensity (See Figure 3). We assume green energy sources emit 23 g-CO2-eq/h and "brown" energy resources averages around 300 g-CO2-eq/h carbon emissions [25]. Based on these initial numbers and the 4% per-year increase in renewable electricity generation, we can estimate the impact of the renewable energy transition on carbon emissions. We estimate that transitioning to green energy resources will reduce the carbon intensity by 14% until 2050, which translates into approximately 0.6% yearly reduction in carbon emissions.

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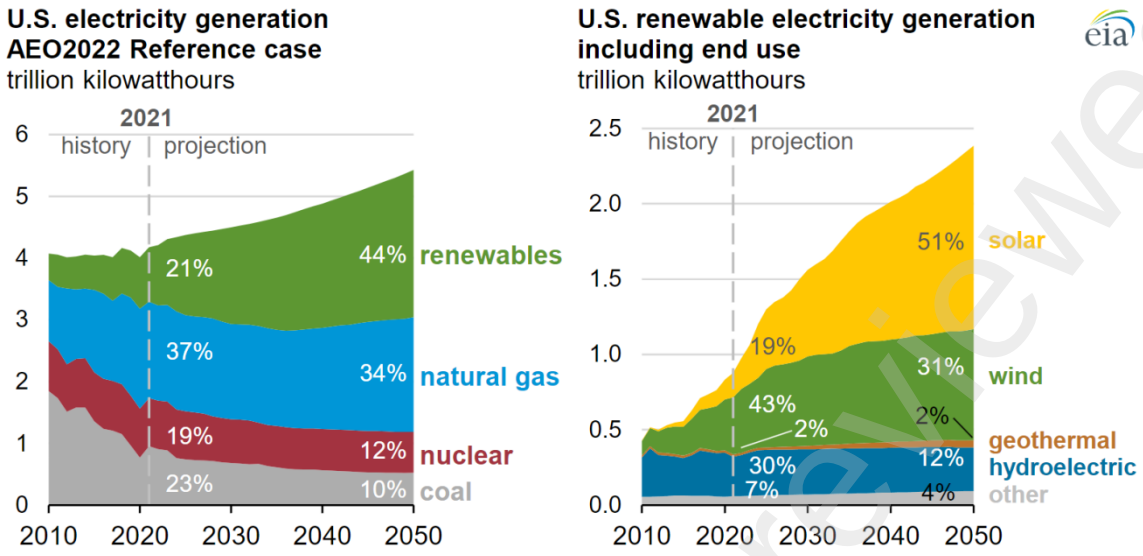

Figure 3: EIA projected renewable electricity generation (EIA AEO2022).

Second, although AI is expected to drive the carbon emissions to significantly higher levels, there are also benefits of using AI for developing sustainability solutions to reduce the carbon emissions. AI-driven sustainability solutions are expected to reduce carbon emissions by 6% to 9% based on various projections. As a conservative approach, we assume AI-driven improvements will be higher during the initial years and will become diminished over time.

4.2. Software-driven Improvements

Recent trends show that AI-based applications, such as training and inference, utilize only around 50% of the available GPU resources, due to bandwidth and communication bottlenecks [26]. These bottlenecks lead to less energy-efficient operation and significantly increase the carbon cost of running AI workloads. With improved software, infrastructure and management techniques, researchers estimate GPU utilization to reach up to 90% [26] (See Figure 4). Based on these projections, we estimate the power efficiency improvements by using the measured power values from recent NVIDIA GPUs, including idle and dynamic power. We assume that the performance of AI applications will scale linearly with increasing utilization and calculate the total energy reduction based on the utilization improvements.

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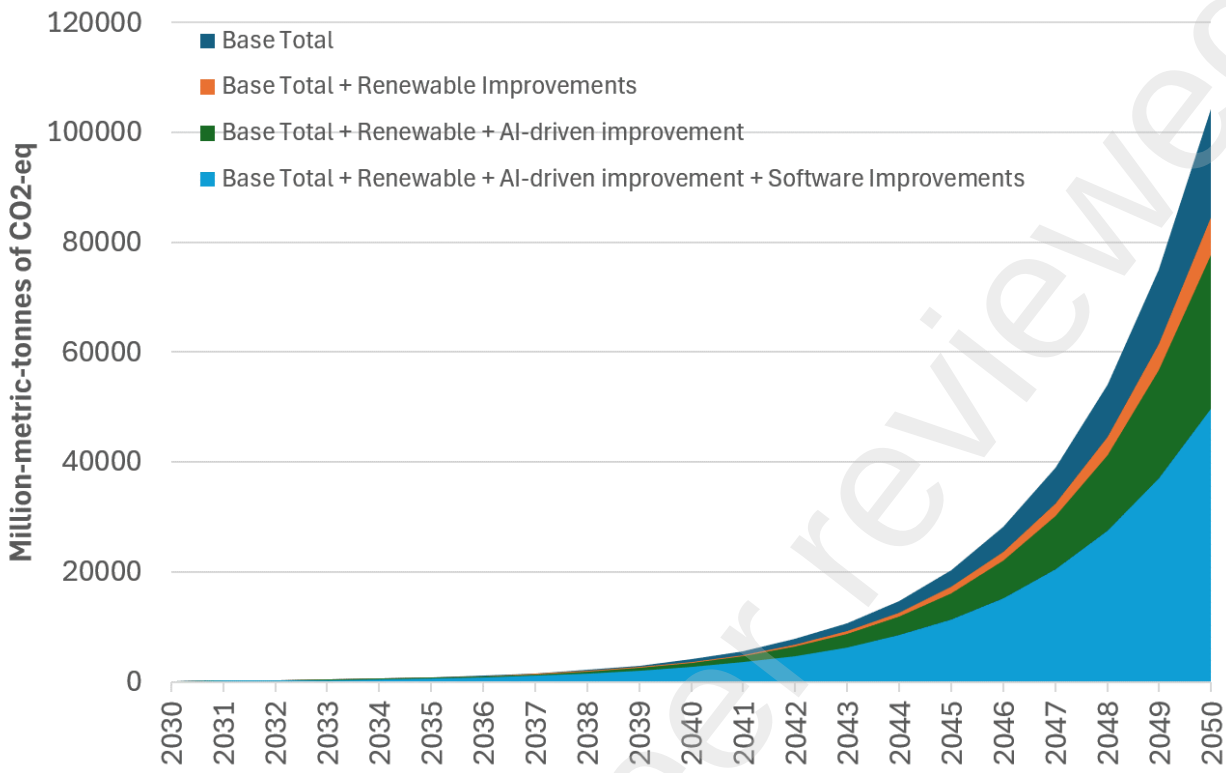

Figure 4: Cumulative big-tech carbon emission projections with and without software improvements.

In Figure 3, we show the cumulative carbon emissions of 4 big-tech companies, Amazon, Google, Meta, Microsoft, from 2030 to 2050. With transition to renewables, improved energy efficiency with the use of AI and software improvements, we project the total carbon emissions can be 48% lower than the baseline case.

4.3. Carbon Offsetting Investments

Big-tech companies are expected to continue investing in carbon offsetting strategies, such as participating in PPAs, buying carbon credits etc. [5]. However, with the growing electricity consumption driven by AI, carbon offsetting costs are expected to increase exponentially. Current carbon credit costs vary between $500-$1000-per-metric ton of CO2 removal, depending on the location and the provider [27]. Based on our carbon emission projections, we calculate the cost of renewable investments to achieve net-zero by 2030, as shown in Figure 5. Amazon will need to invest more than $6-billion dollars to offset their carbon emissions just in 2030. Considering the size of these investments, it is apparent that big-tech companies need to tackle their carbon emissions on various fronts, including better energy efficiency and software improvements.

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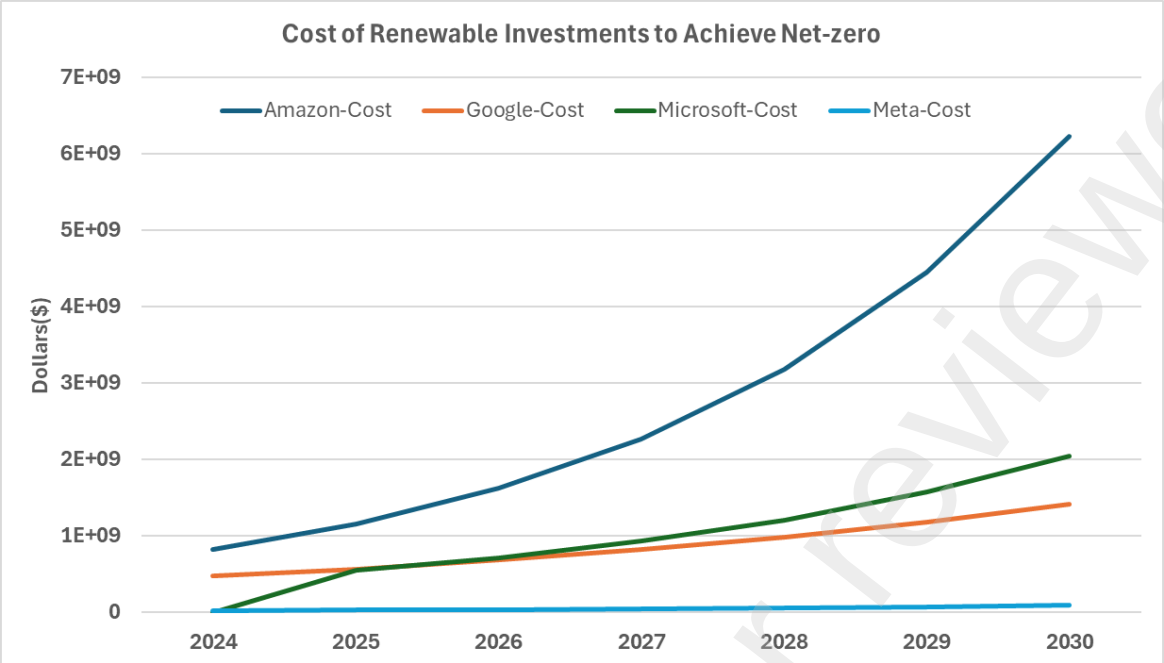

Figure 5: Cost of renewable investments required to achieve net-zero.

5. Conclusion

Achieving net-zero carbon targets for big-tech companies is a challenging task. The need for vast amounts of energy, challenges to switch to renewable sources and the rise of AI make these targets even harder to reach. There is a significant urgency of addressing climate change and there is a great potential for big-tech companies to have a significant positive impact on environment by tackling their carbon emissions. In this article, we show the pressing need for addressing these challenges and the impact of various mitigation strategies. Furthermore, we quantify the impact of AI-driven technologies on the overall data center energy consumption. We believe that through innovation, collaboration, and a commitment to sustainability, big-tech companies can overcome these challenges and play a leading role in the global effort to achieve a net-zero carbon future.

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7. Author Contributions

C.H.: data curation, methodology, conceptualization, visualization, writing.
A.K.C.: supervision, writing.
B.K.S.: funding acquisition, supervision, writing.

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